STEEL FABRICATION

MES-ST-SPC-0001

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REV NO.	REV DATE	DOC NO.	REVIEW CYCLE	REVIEW DATE	PREPARED BY	REVIEWED BY	VERIFIED BY	APPROVED BY (EMPLOYER)
C03	24-JUL-24	MES-ST-SPC-0001	5 Years	24-JUL-29	S Sanjay	2024.07.24 12:50:05 +02'00' TK	2024.07.25 08:14:22 -05'00' eWc	AAG

REVISION HISTORY

REV NO.	REV DATE	PURPOSE	PREPARED BY	REVIEWED BY	VERIFIED BY	APPROVED BY (EMPLOYER)
C01	25-JUL-11	Issued for Use	K H Chong	D Kirkwood	XX	R Covella
C02	14-FEB-18	Issued for Use	FC	FYG	XX	FYG
C03	24-JUL-24	Issued for Use	SS	TK	eWc	AAG

REV NO.	REV DATE	REVISION HISTORY
C01	25-JUL-11	Issued for Use
C02	14-FEB-18	Revised and/or Updated the following • 2 Key Terms and Definitions • 3.2 Standards Materials, to include updates to Structural Steel, Bolts, Welded studs, Steel Deck, Material Testing and Quality Control • 5 Steelwork Details, to include Connections, Floor Grating, and Seal Plates to Tubular Members • 7 Steelwork Fabrication Tolerance and Bolt Holes
C03	24-JUL-24	Revisions made as indicated below: • 2- ASSE changed to ASSP and description changed to 'American Society of Safety Professionals" • 2- Term 'Contractor' is introduced with definition • 3.2.2.1- References to various outdated AISC codes have been deleted • 3.2.2.2- All Descriptions of ASTM standards/Specifications have been prefixed with 'Standard Specifications for …' • 3.2.2.3 – AWS D14.1 and AWS B2.1.1-001 have been added to the list • 3.2.2.5 -ASSE is corrected to ASSP • 3.2.2.6- ISO codes added • 3.2.2.7 – ASME B30.11 is corrected to ASME B30.17 • 3.2.2.8 – PIP standards added • 3.2.3.2- SDI standard for Steel deck on cold formed steel decking added and also added SDI Roof Deck Design Manual • 3.2.3.4- Australian and New Zealand codes added • 3.3- Ma'aden structural Design Criteria added, Deleted standard MD-101-SMPM-PM-QA-PEI-0001 Contractor Quality requirements • 4.1- ASTM A36 added • 4.2.2- F3125/3125M Grade F1852 added to the para on Bolts

		• 4.2.2- Washers mentioned as conforming to 'Type 1' • 4.4 Crane Rail weight modified for ASTM A1 and ASTM A759 conforming rails • 4.5 Conforming code for Gratings modified to A1011/A1011M • 4.8 ASTM A985 is deleted since the code is withdrawn. • 4.9 Reference to Section 7, Type B is deleted. • 4.12 Para added at end – "Employer has the right to inspect all materials and workmanship and shall have unrestricted entry to fabricators' shop at all times while work is being performed. • 5. Completely revised • 5.2 Clause added for supplying of 5% extra quantity for each bolt size/clip and length including nuts and washers. • 5.3 Clause added for provision of drain holes in checkered plates. • 5.6 Clause added for Fall protection system as per latest PIP STE 05501 • 5.10 ASME B30.11 changed to ASME B30.17 • 6 Text inserted regarding responsibility of Fabricator • 7.1 – Code 'ISO 10721-2' added, Clause added regarding commencement of fabrication only on approved drawings • 7.11- Generally revised • 7.12 – Generally revised • 7.14- Text inserted – "Format of Piece mark shall be approved by Employer before usage. • 10.1- Generally revised

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1. INTRODUCTION

1.1 Scope

This document provides the specification for fabrication of all structural steelwork for Ma'aden Projects.

This document is intended to be used in conjunction with the applicable documents listed herein.

1.2 Purpose

The purpose of this specification is to set the minimum engineering standard for all materials, detailing, fabrication, and delivery of structural steelwork.

1.3 Conflicts and Deviations

Any deviation from the requirements stipulated in this document shall go through a waiver approval process as per Ma'aden Capital Investment System (MCIS) guidelines and shall be addressed to Ma'aden's Project Development & Engineering (PDE) team for their review and approval.

2. KEY TERMS AND DEFINITIONS

Contractor	Party Responsible for supplying materials, plant, labor and supervisory services.
Employer	Saudi Arabian Mining Company (Ma'aden) and its authorized representatives and respective affiliates.
Fabricator	The party responsible for furnishing structural and miscellaneous steel

3. REFERENCED DOCUMENTS

This specification requires compliance with the provisions of the latest revision of the relevant standards and codes of the following authorities and technical organizations, as applicable, and specifications which shall be considered to form part of this specification, where applicable. In general, the Contractor shall be responsible for full compliance with applicable Ma'aden Health, Safety and Environmental Standards (MSHEMs). Any deviation from the MSHEMs shall require written approval from the Employer.

3.1 Statutory Regulations

All work performed and materials furnished shall comply with the applicable statutory regulations, codes and other requirements of the Kingdom of Saudi Arabia or local requirements of Project Site. The Contractor shall ensure that he is fully aware and informed of all Government laws, local ordinances, regulations and all orders and decrees of bodies or tribunals having any jurisdiction or

authority over the Project and/or Site, which in any manner affect the Project and those engaged or employed on the Project, or which in any way affect the conduct of the Work. The Contractor shall at all times observe and comply with all such Government and/or Ministry laws, bylaws, ordinances, codes, regulations, orders and decrees.

3.2 Industry Codes and Standards

The work shall conform to the latest adopted revisions of the following applicable codes and standards.

3.2.1 *Local Codes and Standards (for Saudi Arabia)*

All Ma'aden facilities shall comply with latest version of all Saudi Arabia applicable local codes and standards including but not limited to:

Saudi Building Code SBC 306	Steel Structures

For project locations other than Saudi Arabia, the Contractor shall follow the respective applicable local regulations, codes and standards

3.2.2 *International Codes and Standards*

3.2.2.1. OSHA Standards

OSHA 29 CFR Part 1926 Subpart R	Steel Erection

3.2.2.2. American Institute of Steel Construction (AISC)

AISC 303	Code of Standard Practice for Steel Buildings and Bridges
AISC 325	Steel Construction Manual
AISC 360	Specification for Structural Steel Building
AISC 326	Detailing for Steel Construction
ANSI/AISC 341	Seismic Provisions for Structural Steel Buildings
Research Council on Structural Connection-RCSC	Specification for Structural Joints Using High-Strength Bolts

3.2.2.3. American Society for Testing and Materials (ASTM)

ASTM A1	Standard Specification for Carbon Steel Tee Rails

ASTM A6/A6M	Standard Specification for General Requirements for Rolled Structural Steel Bars, Plates, Shapes and Sheet Piling
ASTM A36/A36M	Standard Specification for Carbon Structural Steel
ASTM A53/A53M	Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated Welded and Seamless
ASTM A106/106M	Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service
ASTM A108	Standard Specification for Steel Bars, Carbon and Alloy, Cold-Finished
ASTM A123/A123M	Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
ASTM A143 /A143M	Standard Practice for Safeguarding Against Embrittlement of Hot-Dip Galvanized Structural Steel Products and Procedure for Detecting Embrittlement
ASTM A153/A153M	Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware
ASTM A307	Standard Specification for Carbon Steel Bolts, Studs and Threaded rods, 60,000 psi Tensile Strength
ASTM A384/ A384M	Standard Practice for Safeguarding Against Warpage and Distortion During Hot-Dip Galvanizing of Steel Assemblies
ASTM A385	Standard Practice for Providing High- Quality Zinc Coatings (Hot-Dip)
ASTM A500/A500M	Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes

ASTM A501/A501M	Standard Specification for Hot-Formed Welded and Seamless Carbon Steel Structural Tubing
ASTM A563/A563M	Standard Specification for Carbon and Alloy Steel Nuts (Inch and Metric)
ASTM A572/A572M	Standard Specification for High-Strength Low-Alloy Columbium-Vanadium Structural Steel
ASTM A653/A653M	Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy Coated (Galvannealed) by the Hot-Dip Process
ASTM A759	Standard Specification for Carbon Steel Crane Rails
ASTM A780/A780M	Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings
ASTM A786/A786M	Standard Specification for Hot-Rolled Carbon, Low-Alloy, High Strength Low-Alloy, and Alloy Steel Floor Plates
ASTM A992/A992M	Standard Specification for Structural Steel Shapes
A1011/A1011M	Standard Specification for Steel, Sheet and Strip, Hot Rolled, Carbon, Structural, High Strength Low-Alloy, High Strength Low-Alloy with Improved Formability and Ultra-High Strength
ASTM B69	Standard Specification for Coatings of Zinc Mechanically Deposited on Iron and Steel
ASTM E376	Standard Practice for Measuring Coating Thickness by Magnetic-Field or Eddy Current (Electromagnetic) Test Methods
ASTM E32	Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection
ASTM F436/F436M	Standard Specification for Hardened Steel Washers Inch and Metric Dimensions

ASTM F606/F606M	Standard Test Methods for Determining Mechanical Properties of Externally and Internally Threaded Fasteners, Washers, Direct Tension Indicators and Rivets.
ASTM F844	Standard Specification for Washers, Steel, Plain, (Flat) Unhardened for General Use.
ASTM F959/F959M	Standard Specification for Compressible-Washer- Type Direct Tension Indicators for Use with Structural Fasteners, Inch and Metric Series
ASTM F3125/F3125M	Standard Specification for High Strength Structural Bolts and Assemblies, Steel and Alloy Steel, Heat Treated, Inch Dimensions 120 ksi and 150 ksi Minimum Tensile Strength and Metric Dimensions 830 MPa and 1040 Mpa Minimum Tensile Strength

3.2.2.4. American Welding Society (AWS)

AWS D1.1 /D1.1M	Structural Welding Code – Steel
AWS D1.3 /D1.3M	Structural Welding Code – Sheet Steel
AWS D1.8	Structural Welding Code – Seismic Supplement
AWS QC1	Specification for AWS Certification of Welding Inspectors.
AWS D14.1/14.1M	Specification for Welding of Industrial and Mill Cranes and Other Material Handling Equipment.
AWS B2.1-1-001	Standard Welding procedure Specification (SWPS) for Shielded Metal Arc Welding of Carbon Steel (M-1/P-1, Group 1or 2) 5mm through 19mm thick, E7016 and E7018, in the As-Welded Condition, Primarily Plate and Structural applications

3.2.2.5. Architectural Metal Products (AMP)

AMP 510	Metal Stairs Manual, Fifth edition

3.2.2.6. American Petroleum Institute (API)

API 650	Welded Steel Tanks for Oil storage

3.2.2.7. American Society of Safety Professionals (ASSP)

ANSI/ASSE A1264.1	Safety Requirements for Workplace Walking/Working Surfaces and Their Access; Workplace, Floor, Wall and Roof Openings; Stairs and Guardrails/Handrail Systems.
ANSI/ASSE Z359	Fall Protection Code and Arrest Standards Package

3.2.2.8. International Standard (ISO)

ISO 17607-3	Steel Structures – Part 2: Fabrication and Erection

3.2.2.9. American Society of Mechanical Engineers (ASME)

ASME A120.1	Safety Requirements for Powered Platforms and Traveling Ladders and Gantries for Building Maintenance
ASME B30.17	Cranes and Monorails (With Underhung Trolley or Bridge)

3.2.2.10. Process Industry Practice (PIP)

PIP STS05130	Structural and Miscellaneous Steel Erection Specification
PIP STS05120	Structural and Miscellaneous Steel Fabrication Specification
PIP STE05501	Fixed Ladders Design Guide
PIP STF05501	Fixed Ladders Fabrication Details
PIP STE05530	Grating Design Guide
PIP STF05511	Standard Stairs Fabrication Details
PIP STF05520	Pipe Guards and Handrails Fabrication Details
PIP STF05521	Angle Guards and Handrails Fabrication Details
PIP STF05530	Grating Fabrication Details

3.2.3 *Alternative Design Codes*

3.2.3.1 National Association of Architectural Metals Manufacturers (NAAMM)

NAAMM MBG 531	Metal Bar Grating Manual

3.2.3.2. Steel Deck Institute

Engineering Manual No DDM4	Diaphragm Design Manual
Engineering Manual No FDDM2	Floor Deck Design Manual
Engineering Manual No RDDM2	SDI Roof Deck Design Manual
Engineering Manual No CFDDM	Steel Deck on Cold Formed Steel Framing Design Manual
SDI	Manual of Construction with Steel Deck- No. MOC3
SDI	Standard Practice Details-No SPD2
ANSI/SDI/COSP 2023	Code of Standard Practice for Steel Deck

3.2.3.3. Steel Joist Institute

SJI	Standards, Specifications and Load Tables

3.2.3.4. Australian/New Zealand Standard (AS/NZS)

AS/NZS 1554.1	Structural Steel Welding Part1: Welding of Steel Structures

3.3 Ma'aden Standards

MD-101-SMEM-EG-ST-SPC-0002	Steelwork Erection
MD-101-SMEM-EG-ST-SPC-0009	Protective Coating for Steelwork
MD-101-SMEM-EG-ST-CRT-0001	Structural Design Criteria-Buildings & Structures
MD-101-SMEM-EG-ST-DST-0001 to 0029	Ma'aden Standard Drawings

4. MATERIALS

4.1 Structural Steel

All material shall be new, free from rust, pitting, flaws, cracks, and any other defects and shall have a smooth uniform finish. All steel plates, flats and rolled sections in the work shall be in accordance with ASTM A36/A36M, ASTM A6/A6M, ASTM A572/572M or A992/A992M as appropriate.

Hollow Structural Sections (HSS) shall conform to ASTM A500/A500M Grade B or A501/A501M Grade A, unless otherwise specified.

Pipe Sections shall be ASTM A53/A53M Type E or S, Grade B, or ASTM A106/A106M Grade B

The steel supplier shall be subjected to an approval and conformity process by the Employer. Supplier shall submit a pre-qualification document, in which Employer shall evaluate, make inspection on the plant and facilities, evaluate the conformance assurances and certificates, prior to approval.

Mill certificates and heat numbers shall be supplied for all steel used in the works, cross-referenced to the item numbers to be cut from the materials, prior to the commencement of cutting or fabricating. The certification shall demonstrate the compliance of the steel with this Specification.

Unidentified steel sections shall not be used in any part of the work. Only steel sections that comply with the requirements of this Specification shall be used in the work.

4.2 Bolts

4.2.1 *General*

Boltheads and nuts shall be hexagonal except where otherwise specified on the drawings. Threads and machined surfaces shall be greased. Nuts shall be capable of being threaded the full length of the bolt by hand.

4.2.2 *High Strength Bolts*

Bolts shall conform to ASTM F3125/3125M Grade A325 Type 1.

Washers shall conform to ASTM F436/F436M Type 1.

Washers for use with short slotted or oversized holes or over 1 -inch diameter A490 bolts must also be 5/16 of an inch or thicker.

Direct tension indicator (DTI) washers shall conform to ASTM F959/F959M, unless otherwise specified. Heavy Hex nuts shall conform to ASTM A563/A563M Grade 10S.

4.2.3 *Standard Bolts*

Bolts shall conform to ASTM A307 Grade A heavy hex.

Washers shall conform to ASTM F436/F436M Type 1.

Washers for use with short slotted or oversized holes or over 24 mm diameter A490 bolts must also be 8 mm or thicker.

Heavy hex nuts shall conform to ASTM A563/A563M Grade A.

4.3 Weld

Welding filler metal shall conform to AWS D1.1, low hydrogen with an electrode strength of 400 MPa minimum yield strength and 480 MPa minimum tensile strength. However, E60XX electrodes may be used for tack welding.

4.4 Crane Rail

Crane rails shall be as follows unless otherwise specified in this Specification:

- ASTM A1 for rails weighing 29.8 kg/m to 41.7 kg/m
- ASTM A759 for rails weighing 51.6 kg/m to 87.7 kg/m.

4.5 Floor Grating

Grating shall conform to ASTM A1011/A1011M or A36/A36M, galvanized in accordance with ASTM A123 and NAAMM MBG 531. Top surface of bearing bars shall be plain unless serrated is specified. Grating for platforms, landings and floor shall be welded Type 30-102 (32X4.8) with bearing bars 30mm center to center, as manufactured by Fisher or Ludlow or approved equal.

4.6 Grating Clamps

Grating clamps shall be reusable type, as manufactured by Hilti, or approved equal. Entire assembly to be painted or stainless steel. Clamps shall be of size required for the thickness of grating. Grating clamps shall be reusable type, as manufactured by Hilti, or approved equal. Entire assembly to be hot-dip galvanized or stainless steel. Clamps shall be of the size required for the thickness of grating.

Grating clamps shall be reusable type, as manufactured by Hilti, or approved equal. Entire assembly to be painted or stainless steel. Clamps shall be of the size required for the thickness of grating.

4.7 Floor Plate

Checkered plate shall conform to ASTM A786/A786M Pattern 4 or Pattern 5 and ASTM A36/A36M unless otherwise specified.

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4.8 Stair Tread

Grating stair treads shall conform the same standard as noted in Section 4.5. Alternative stair tread shall be subject to approval by the Employer. Stair treads shall have abrasive or checkered plate nosing unless otherwise specified.

4.9 Handrail

Pipe and Angle Hand railing shall comply to the requirements of PIP STF05520 and PIP STF05521 respectively.

4.10 Welded studs

Headed Welded studs shall conform to ASTM A108 Grade 1010 through Grade 1020 and AWS D1.1 Section 7, Type B.

4.11 Steel Deck

Steel deck shall be in accordance with SDI Design Manuals No DDM4, No FDDM2, No RDDM2 and No. CFDDM unless otherwise specified.

Roof Deck

Roof deck shall conform to ASTM A653/A653M, Grade A, having a minimum yield strength of 227.5 Mpa (33,000 psi) and a minimum thickness of 0.76 mm (22 gage). Decking shall receive a commercial zinc coating G90 in accordance with ASTM A653. For more details on other types of roof decks, material, coating etc. refer to design drawings and Architectural specifications.

Form Deck

Form deck shall be manufactured from steel conforming to ASTM A653/A653M, Grade A, having a minimum yield strength of 227.5 MPa (33,000 psi) and a minimum thickness of 0.91 mm (20 gauge) or as noted on design drawings. Decking shall receive a commercial zinc coating (G90) in accordance with ASTM A653/A653M. All form deck to be supplied as composite deck.

Form deck shall resist construction loads including the weight of fresh concrete. After the concrete hardens, the steel deck and the concrete shall be interlocked. The steel deck shall provide the means for this interlocking. Panel configuration shall be in accordance with SDI and as shown on the drawings.

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The deflection (simple span) of the decking under the weight of fresh concrete shall be limited to L/240.

4.12 Steel Joists

Steel joists shall be in accordance with SJI Standard Specifications and Load Tables, unless otherwise specified.

4.13 Material Testing and Quality Control

Each piece of mill material shall be legibly marked with the heat number, size of section, length and mill identification marks in accordance with ASTM A6/A6M plus the fabrication mill order number.

The Contractor shall have a written material handling procedure, quality control program and inspection procedures document that shall provide details of how compliance with the requirements in this specification and the design drawings shall be achieved.

Welding procedures and individual welders shall be qualified in accordance with the requirements of AWS D1.1. Contractor's inspectors shall be qualified and certified as AWS Certified Welding Inspectors in accordance with the provisions of AWS D1.1 or AWS QC1 or shall be trained by and working under the supervision of an AWS Certified Welding Inspector.

The Contractor shall be solely responsible for quality control of all materials and workmanship including the following:

- Coordination with the Employer, including works associated with the handling and re-stacking of materials, for inspection and testing.
- Replacement of materials which have been rejected as a result of testing.
- Testing of steel quality from all sources.
- Retesting, if directed by the Employer.
- Procedure to establish welding procedures and ensure welding quality. The NDT requirements are:
    - Visual inspection of 100% of all welds shall be performed.
    - Ultrasonic tests examination to AWS D1.1 shall be carried out on a minimum of 10% of complete penetration and partial penetration butt welds.

- Ultrasonic tests examination to AWS D1.1 shall be carried out on 100% of complete penetration welded splice joints.
- Magnetic particle testing to AWS D1.1 shall be carried out on a minimum of 10% of all other weld types.

The Employer may approve in some circumstances the substitution dye penetrant to AWS D1.1 in lieu of magnetic particle testing.

All non-destructive testing, for which the Contractor is responsible, shall be undertaken by an organization registered with the ASTM or any other approved international testing authority.

The Contractor shall be responsible for any chemical analysis and mechanical testing, prior to fabrication, as required by the Employer, if the certification supplied does not demonstrate the compliance of the steel with this Specification. Testing shall be undertaken by an authorized laboratory. Results of tests shall be presented in a manner that demonstrates compliance with the requirements of relevant American Standard for the type of material tested.

The Employer has the right to inspect all materials and workmanship and shall have unrestricted entry to the fabricator's shop at all times while work is being performed.

5. STEELWORK DETAILS

All field, galvanized or coated connections shall be bolted, and all shop connections shall be either bolted or welded unless specified in design drawings.

5.1 Connections

5.1.1 *General*

Connections shall be designed and detailed in accordance with AISC, the design drawings and this Specification.

Design, detailing and fabrication of welded connections shall be in accordance with either the AISC Allowable Stress Design or Load and Resistance Factor Design Method as noted in Contract Documents.

Shop welded and shop bolted connections shall be so located and detailed as to allow parts of the structures to be safely assembled in the shop into the largest components consistent with practical limitation determined by transportation and constructability.

All fittings shall be made from a minimum of 10 mm thick plate, unless noted otherwise on the design drawings.

Beam-to-column connections shall be so arranged as to facilitate erection without the tilting of columns.

Shop fabricated connections shall be welded whenever possible or high strength bolted bearing type connections (Type N) using A325 bolts. Unless otherwise noted on the design drawings or specifications, all connections not fully detailed on the design drawings shall be designed or selected by the Fabricator.

5.1.2 *Fabricator designed Connections*

All connections not fully detailed on the design drawings shall be designed and detailed by the fabricator.

1. The minimum design capacity of all frame-beam connections shall be the member end reaction shown on the design drawings. When the member end reaction is not shown, the minimum design capacity shall be as per Clause 8.5.8 of Ma'aden Standard MD-101-SMEM-EG-ST-CRT-0001 unless otherwise specified in the Contract Documents.
2. The Fabricator shall have a professional engineer design or supervise the design of all Fabricator-designed connections.
3. Fabricator-designed connections shall meet all requirements of the contract documents. In all Cases the design shall consider the entire joint (including beams, girders, columns, and bracing) and shall take into account bolt shear, combined bolt tension and shear, prying action, local bending, coped beam capacities, block shear, web buckling, etc.
   Fabricator shall design and furnish stiffeners for column webs, column flanges, and elements of other members involved as required.
4. Sample engineering calculations for each type of the fabricator-designed connection shall be sealed and signed by the responsible professional engineer who designed or supervised the design of the connections. Additionally, when required by the contract documents, all shop drawings containing fabricator-designed connections shall be sealed and signed by the responsible professional engineer. Shop drawings containing fabricator-designed connections shall be sealed and signed by the responsible professional engineer.
5. Review of shop drawings detailing fabricator-designed connections and engineering calculation sheets by the Employer does not relieve the fabricator of responsibility for both the design adequacy and detailing of connections designed by the fabricator.

5.2 Floor Grating

Floor grating shall comply with the requirements as noted in Section 4.5.

Grating shall be removable unless otherwise specified on the design drawings.

The clear span of the grating between supporting beams shall not exceed 1200 mm.

All grating panels shall have welded closed ends with end banding bars having the same size as main bars.

The distance between grating panels shall be nominally 10 mm and shall be a maximum of 20 mm. The clearance between the grating panel and toe plate of hand railing shall be 10 mm.

Openings in grating, to permit pipework, ducting, equipment, structural columns, and other such components to be positioned or to pass through, shall be designed as follows:

1. Where the diameter or longer dimension of opening exceeds 200 mm, the grating shall be split into two panels along the center line of the to facilitate installation upon completion of piping work.
2. Openings shall be provided with banding bars of the same size as the main bars except where stiffening ring reinforcement is used in lieu.
3. A maximum of 20 mm shall be provided between the face of banding bar around opening and bare or fireproofed face of members to be positioned.
4. Sufficient grating clips shall be supplied with grating to meet the following requirements:
    - Grating shall be secured to the supporting steel beams by removable grating clamps. A minimum of four clips per panel shall be used.
    - Clips to be installed at 1200 mm maximum intervals in direction at right angles to load bar along edge and intermediate supports, with a minimum of two clips per support.
    - The clips shall be installed at 500 mm maximum intervals in direction at right angles to load bar along edge and intermediate supports, with a minimum of two clips per support.
    - All clip bolts shall be supplied with a galvanized spring washer above the top plate. A minimum of 5% extra quantities of each bolt size/Clips and length, including nuts and washers shall be provided for field erection.

Removable grating panels provided for the purposes of operations and maintenance shall weigh a maximum of 20 kg and shall be fixed with a minimum of four fastening clips.

Where the floor slope is greater than or equal to 12°, the floor grating shall have 10 mm square bars at 300 mm centers welded on the top surface at right angles to the slope. Top edges of bars shall be chamfered.

Joints in floor panels perpendicular to the span of grating shall only occur over support members.

Notching of bearing bars at supports to maintain elevation is not permitted.

Provide a minimum of 1" bearing for grating at supporting members.

Openings dimensioned on the design drawing shall be cut and banded in the shop. Un-dimensioned openings shall be cut in the field.

5.3 Floor Plate

Floor plating shall comply with the requirements as noted in Section 4.7.

Chequered plate shall be 6 mm thick minimum steel plate characterized by a raised angular pattern on top surface. Unless noted otherwise, chequered plate shall be coated in accordance with the Project Specification, MD-101-SMEM-EG-ST-SPC-0009, Protective Coating for Steelwork. Chequered plates shall have ½ inch (12mm) diameter drain holes provided for each 20 square feet (2 Square meters) of area, with a minimum of one hole per panel.

5.4 Handrail

Hand railing shall comply to the requirements in Section 4.9.

Hand railing shall be supplied in standardized, prefabricated panels in the largest components consistent with transportation and constructability. The panels shall be supplied complete with all associated bends and closures, so that site work for installing the hand railing is minimized.

Railing (Process Buildings/Structures)

Fabricate from pipe as shown on design drawing, with closed ends, heated and bent smoothly without distortion. Return ends of wall mounted rails to wall. Provide smooth continuous transition around turns and bends. Miter, weld, and grind smooth where exposed.

Load

Concentrated 200-pound load applied in any direction at any point on the railing.

Toe Plate

Provide bar 4 inches by 1/4 of an inch at platform handrails

5.5 Stairs

Stairs shall comply with the requirements of ASSE A1264.1.

Stairs shall have bolted treads and shall be pre-assembled into sections for transport. Stiffening bracing for stairs shall be provided at undersides of stringers where a stair has more than 18 rises without horizontal support.

5.6 Ladders

Permanent Travelling Ladders shall comply with the requirements of ASME A120.1.

Ladders may be oriented vertically where space is limited otherwise the provisions of ASME A120.1 shall apply. Components of ladders, i.e. stiles, rungs and cages shall be shop welded, including cage-to-style connections.

Fall protection systems to be provided as per latest PIP STE 05501 –"Fixed ladders Design Guide ". Ladders on insulated vessels shall be detailed to allow adequate toe clearance after installation of insulation and cladding. All connections to vessels shall be external to the insulation.

5.7 Seal Plates to Tubular Members

All tubular members shall be seal welded to prevent the access of moisture to the inside of the members. The minimum thickness of sealing plates shall be 6 mm. Galvanized tube members with end plates require weep holes for the galvanizing procedure.

5.8 Temporary Members

Temporary bracing and supports for transportation, where required, shall be designed with bolted connections. Wherever possible, anchor bolts holes and other bolt holes detailed for field bolting shall be used for connection of the temporary supports.

5.9 Crane Beams

Crane rails shall comply with the requirements as noted in Section 4.4.

Crane rails on each runway beam shall be detailed with joints longitudinally staggered. Rails shall not be spliced at the same point as crane runway beam splices. Expansion joints, where required, shall be provided in accordance with design drawings.

5.10 Monorail Beams

Monorail beams for monorail cranes or trolleys shall be in accordance with ASME B30.17.

Monorail beams shall be finished in a painted topcoat. The color shall be 'Golden Yellow', Color Y14 to ASME B30.17 (RAL Equivalent 1004). The rated safe load capacity shall be painted on the web each side of the beam, e.g. "2 Ton WLL". The lettering shall be black with a white background, at least 100 mm high and located so as to be visible from the operating platform.

6. SHOP DETAILING

The Contractor shall be responsible for the production and checking of shop detail models, drawings, marking plans and other specified deliverables. Errors or omissions in the shop detailing deliverables shall be the Contractor's responsibility. The approval by the Employer of shop drawings does not relieve Fabricator of responsibility of correctness of shop drawings.

For each piece of structural steel to be painted the shop detail drawings shall detail the system required and the areas to be left untreated. The paint system shall conform to standard specification as referenced in Section 8.2.

The Contractor shall schedule the work to include the production of shop detail deliverables.

Shop drawings and erection drawings shall be prepared in accordance with the AISC Standards listed in this specification unless otherwise specified in contract documents, state or local laws or building codes.

Erection drawings shall reference the corresponding design drawings; shop drawings shall reference the corresponding erection drawings.

Shop drawings shall clearly show the specification and grade of steel to be used.

Erection and shop drawings shall be grouped in sets and identified separately for each building, structure or yard area.

Erection drawings shall clearly show the mark number and position for each member.

The purchase order number shall be shown on all erection and shop drawings.

Shop drawings shall state the welding electrode to be used.

Surface preparation and shop-applied coatings, including areas to be masked, shall be noted on the shop drawings.

The Contractor shall provide a bolt list and a list of other fasteners showing the number, grade, size and length of field bolts and other fasteners for each connection. These lists may be shown on either the shop drawings or separate sheets.

In the event that drawing revisions are necessary, Contractor shall clearly flag onto the shop drawings all changes showing the latest revisions.

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7. STEELWORK FABRICATION

7.1 General

Fabrication shall comply with this Specification, standard drawings, design drawings and the applicable codes, in particular AISC 326, ISO 17607-3 and SBC 306, Section 13.2.

Fabrication is allowed to start only after approval of the shop drawings by Employer.

7.2 Steel Certificates

The Contractor shall supply manufacturer's test certificates stating chemical analysis and mechanical tests for every batch of steel used by it in the project.

The Employer reserves the right to call for certificates at any time for any batch of steel. The Contractor may be required to submit samples to Employer approved laboratory for testing. The cost of this shall be borne by the Contractor.

Where testing is specified on the drawings or elsewhere, the Contractor shall engage an Employer approved testing authority to test all plate of 20 mm or over for laminations or impurities. The method shall be ultrasonic testing, unless otherwise directed.

7.3 Steel Supplier

The Contractor shall nominate the steel supplier for the Employer's approval. Upon acceptance of the steel Contractor, the Contractor shall not obtain steel supplies from any other source for the duration of that Contract without written approval from the Employer. All steel material shall be new, undamaged, and free from rust, foreign material and surface imperfections. Steel to be stored shall be stacked free from contact with the ground in a manner to avoid collection of water, deformation, and overstressing.

7.4 Variation from Drawings

No departure from sizes, dimensions, quantities, or materials shown on the drawings shall be permitted unless prior written approval is obtained from the Employer.

7.5 Shop Errors

Material improperly detailed or wrongly fabricated by the Contractor itself, so that its erection in the field necessitates extra work, shall be the responsibility of the Contractor, who shall cover the entire cost of the correction of shop errors, the replacement of wrongly fabricated materials and all associated costs incurred due to delays or additional work effected by other Contractors.

The Contractor shall be responsible for the correction of shop errors and all associated costs as described above.

7.6 Straightness, Cambering and Curving

It is the Contractor's responsibility to provide fabricated steelwork in accordance with the dimensional tolerances as per AISC 303.

Any straightening of material or bending to produce curvilinear forms shown on the drawings shall be carried out by methods that shall not damage the material and shall be in accordance with a methodology approved by the Employer.

Local application of heat or mechanical means is permitted to be used to introduce or correct camber, curvature and straightness. For hot rolled structural shapes, hollow structural sections (HSS), plates and bars conforming to the standard designations listed in Section A3.1a, the temperature of heated regions shall not exceed 1,200°F (650°C), except that for ASTM A514/A514M the temperature of heated regions shall not exceed 1,100°F (590°C) and for ASTM A709/ A709M, ASTM A913/ A913M and ASTM A1066/ A1066M, the temperature shall not exceed the maximum as specified in the corresponding ASTM material standard. Subject to the approval of the engineer of record, alternative temperature limitations in accordance with recommendations by the producer of the material shall be used.

The deviation of crane girder straightness (sweep and camber) shall not exceed 10 mm, plus or minus, with a maximum rate of change of 6 mm in 20 feet.

7.7 Cutting

Cold sawing shall be used where sections are required to be cut accurately to length, or where a rough edge may spoil the aesthetics of the structure. Columns shall be cold sawn, or machine cut to provide a full bearing contact at the base plate or at the column splice, in accordance with SBC 306 Section 13 tolerance requirements for a bearing interface.

Plates thicker than 6 mm, whose edges are to be welded, shall not be shear cut. At other times machine gas cutting, cropping and shearing will be accepted.

Defects shall be removed from sheared, or flame cut plates, or ends of rolled weld metal, etc.

All free edges shall be smooth and suitably dressed without distortion.

Miters shall be dressed smooth and neatly fitted.

At internal corners of notches in steelwork, a radiused profile shall be provided as specified to a minimum of 11 mm.

For thermal cutting, beam coping, weld access holes, finish of base plates, holes for anchor bolts refer to AISC A360-22 Chapter M.

7.8 Dressing

All sharp corners, rough edges, burrs, weld splatters and the like shall be removed by grinding or other suitable means prior to applying any surface treatment.

7.9 Tolerance

Dimensional tolerances shall be in accordance with the AISC Code of Standard Practice.

7.10 Bolt Holes

Where slotted holes are specified or shown, such holes shall be accurately shaped with sides smooth, parallel and in accordance with AISC and SBC 306, Section 10 'Splices'.

7.11 Splices

All members shall, where applicable, be of a single length of material.

A splice in any member shall only be carried out with the approval of the Employer and at the location nominated. All splices shall develop full strength of the member unless noted on the drawings.

Welded splices in compound members shall be staggered and the component plates and sections spliced before compounding. The minimum length of any steel piece in a welded splice shall be 1m.

Critical field splices and joints shall be mated in the shop on completion of fabrication to ensure satisfactory fit when installed.

All column splices shall be field-bolted and shall conform to AISC Detailing for Steel Constructions.

Beam-splice connections are not permitted unless shown on the design drawings.

Design column base plate connection as welded and column field splices as bolted unless.

noted otherwise. The splice location shall be approved by Employer's Representative.

Columns shall be 1-piece, without splices, except as otherwise shown on the drawings. Splices desired for convenience of Supplier shall be subject to acceptance by Employer's – Representative.

Top and bottom chords of trusses and sway frames shall be 1-piece, without splices, except as may otherwise be shown on the drawings. Splices desired for the convenience of Supplier shall be subject to acceptance by Employer's Representative.

7.12 Connections General

Connection design shall conform to the standards referenced in Section 3.2.2.1.

A minimum of two bolts shall be used for all connections unless shown otherwise on the design drawings.

Where a beam connection is required to carry moment, design the connection for 100 percent of the moment capacity of the smaller connected member, as a minimum.

X-bracing: Design x-bracing as tension bracing. Design connections for 50 percent of the allowable tension capacity of the member.

K-bracing: Design k-bracing and single bracing members in a structure level as compression bracing. Design connections for 100 percent of the allowable compression capacity of the member.

Crane rail joints shall be staggered at least 3.7m (12 feet) on opposite sides of the runway and shall be at least 0.9m (3 feet) from crane girder joints.

7.12.1 *Welded Connections General*

Welded connection design shall conform to AWS D1.1, AISC 360 and AISC Detailing for Steel Construction.

The minimum fillet weld size is 6 mm for structural welds. Seal welds may be 3 mm minimum fillet

weld.

Welds shall be continuous, unless otherwise approved.

The contractor shall design and install erection clips for field-welded connections.

The contractor shall remove all run-off bars and extension tabs.

7.12.2 *Qualification of Welding Procedures*

All welding procedures shall be qualified in accordance with AWS D1.1.

All welding procedure sheets and procedure qualification records of testing shall be submitted to the Employer for approval for each consumable, preparation, or any changes in essential variables. Production welding shall not commence before receipt of this approval in writing.

Tests for qualification of welding procedures shall be in accordance with AWS D1.1, except where suitable prior experience with similar welding procedures cannot be demonstrated. In this case the Employer may elect to increase the amount of testing required for the qualification. AS 1554.1 can be used as an alternative code, subject to Employer's approval.

7.12.3 *Qualification of Welding Personnel*

Welding Supervisors shall be qualified in accordance with AWS QC1. The Employer may disqualify any welder who does not produce welds in accordance with the approved welding procedure. Irrespective of the above document submissions, it is the Contractor's responsibility to ensure the end product conforms with the specifications.

7.12.4 *Cleaning Welds*

Before welding over previously deposited weld metal, all traces of slag shall be removed from the deposit by chipping, and the deposit and adjoining base metal shall be wire brushed until clean at all points. This requirement shall apply not only to successive layers, but also to successive beads and to the overlapping area wherever a junction is made on starting a new electrode.

All completed welds shall be cleaned of all slag by mechanical wire brush or other suitable means until clean at all joints, in conformance with AWS D1.1.

7.12.5 *Repairs & Distortion*

Welds shall be repaired in accordance with SBC 306, Section 14 except that flame gouging is not permitted. Repairs to welds shall be re-inspected by the original NDT method used to detect the defect.

All repair welding, inspection and testing shall be fully recorded and documented.

Repairs shall not commence before receipt of approval in writing from the Employer of the corrective procedure.

Repair, chipping, or grinding of welds shall be done without any gouging, grooving, or reduction in the base metal thickness.

The method of sequence of welding shall be performed so that the heat of welding is distributed to minimize shrinkage stresses and distortion.

Members distorted by welding shall be straightened by mechanical means or by carefully controlled application of a limited amount of localized heat. The temperature of the heated areas shall not exceed 600°C as measured by thermo-melt crayon or other suitable means.

Before any attempt is made to straighten or correct distorted steel, a detailed procedure or work method shall be submitted to the Employer for approval. The procedure shall demonstrate that the method will not reduce the properties of the steel below those specified. The use of water or other methods to accelerate the rate of cooling shall require prior approval by the Employer.

7.13 Bolted Connections

Connection design shall be in accordance with RCSC Specification for Structural Joints Using High - Strength Bolts, AISC 325 and AISC 326.

Unless otherwise specified in design documents, ¾ inch (M20) diameter ASTM F3125/F3125M-A325 Type 1, high strength bolts shall be used in all bolted structural connections. ASTM A307 bolts 5/8 inch (16mm) or smaller diameter shall be permitted for ladder, stair tread, purlin, girt, door frame and guard and handrail connections. However, larger ASTM F3125/AF3125M bolts may be used when required to meet connection capacity requirements.

ASTM A307 bolts larger than 5/8 inch (16mm) shall not be permitted. Slip-critical type connections A325-SC shall be used if noted in contract documents.

All bolt lengths shall be determined in accordance with the commentary on RCSC Specifications for Structural Joints Using High Strength bolts, Table C-2.2.

Provide additional bolts or connection devices where required to comply with OSHA 29 CFR Part 1926, Subpart R -Steel Erection.

High-strength bolt connections shall be bearing type with threads included in the shear plane (type A325-N).

Slotted or oversized hole dimensions shall be in accordance with Table J3.3 of the ANSI/AISC 360-16.

For Standard holes a minimum of one hardened washer shall be supplied with each bolt.

Hardened washers shall be provided under all boltheads and/or nuts adjacent to any ply with oversized or slotted holes. For standard holes, a minimum of one hardened washer shall be supplied with each bolt.

If DTI is to be used, it shall be installed according to the manufacturer's published specifications.

All Bolts shall be supplied galvanized, or zinc coated, as per ASTM A153.

A minimum of 5% extra quantity of each bolt size and length, including nuts and washers, shall be furnished by Contractor for field erection.

Mechanically galvanized bolts or nuts shall not be interchanged with hot-dipped galvanizing nuts or bolts, respectively.

Wherever possible, bolts lengths shall be nominated as multiples of 10 mm.

After tightening a bolt at least one thread shall project beyond the nut.

Members with shop-bolted connections shall be assembled to the same tolerances as shop-welded members.

High strength bolted connections shall be indicated clearly on the erection drawings.

Steel components designated as removable shall be marked as such, and their connections shall be detailed accordingly.

7.14 Construction with Hollow Sections

Where painting is specified as the corrosion protection system, the Contractor shall fully seal all hollow sections.

Welded joints in hollow sections shall develop the equivalent full strength of the member.

7.15 Identification

Each piece of structural steel shall be given a piece mark. Piece marks shall be shown on erection diagrams and shop detail drawings including marking plans.

Each piece mark shall be prefixed by the Area Number. Facility prefix numbers shall be an integral part of piece marks and shall be shown on all erection diagrams and shop detail drawings. Format of Piece mark shall be approved by Employer before usage.

Piece marks shall be hard stamped (metal stamped) in letters and numbers with a minimum height of 12 mm and highlighted with a conspicuous paint flash. The metal stamping shall be of a depth sufficient to preserve identification throughout painting including blasting.

An alternative identification system such as bar-coding may be submitted for approval, providing it complies with the first two paragraphs.

Piece marks shall be located as follows:

- Beams shall be marked on the top face within 200 mm of the north or west end.
- Columns shall be marked on the north face (flange or web, as applicable) within 200 mm of the base plate.
- Horizontal bracing shall be marked within 200 mm of the north or west end.
- Vertical bracing shall be marked within 200 mm of the top end.
- Plates, frames and like not listed above shall be marked in an easily accessible and obvious position and such location shall be shown on the shop detail drawings.

Small loose pieces shall be securely bagged or bundled and clearly marked. Bolts, nuts, and washers shall be provided in secure containers, partitioned, and clearly marked for each size.

8. PROTECTIVE COATING FOR STEELWORK

8.1 Galvanizing

8.1.1 *General*

The surface preparation and protective coating for all structural steelworks shall be in accordance with Project Specification, MD-101-SMEM-EG-ST-SPC-0009, Protective Coating for Steelwork.

8.1.2 *Repair of Galvanized Steel*

Any damage to galvanizing shall be repaired in accordance with ASTM A780/A780M.

Before repair of damaged galvanizing coating, the exposed substrate metal shall be cleaned to bright metal and free of all visual rust, oil or grease. Any non-adhering galvanizing shall be removed to the extent that the surrounding galvanizing is integral and adherent.

When surface defects exceed 2% of a member area, the defects shall be repaired by re-dipping the member in the zinc bath.

Cold repair using an organic zinc-rich coating is allowed if the total damaged area is less than 1% of the total coated area of the member being repaired and no single repair is greater than 1300 mm2 or 300 mm long. The dry film thickness shall be 0.05 mm to 0.08 mm and contain a minimum of 65% zinc dust by weight.

Hot repairs shall be made in the shop if any of the following conditions exists:

a) Total damaged area is greater than 1%, but less than 2%, of the total coated area of the member being repaired.

b) Any single repair is at least 1300 mm2 in area.

c) Any single repair is 300 mm long or more.

Hot repairs shall be made using zinc alloy rod or powder manufactured for the repair of galvanized steel.

Flux, heavy ash or heavy dross inclusions shall be removed by brushing, grinding or filling as required.

Galvanized steel that has been rejected shall be stripped, re-galvanized and submitted again for inspection. Contractor shall be responsible to avoid warpage and embrittlement during galvanizing.

Correction of excessive warpage, that exceeds applicable standard for the sections used, shall be by press straightening when possible. The application of localized heating to straighten must be approved.

If galvanized tension control bolts are used, all bare steel surfaces (i.e. bolt ends) shall be repair galvanized as per this section.

8.2 Unpainted Surfaces

The contact surfaces of all bolted joints designated on the design drawings shall be left uncoated. The Contractor shall clearly identify the surfaces to be left uncoated.

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Structural steel shall remain unpainted in the following areas:

- Where concrete and steel are in contact
- At contact areas to be field welded
- Crane rails
- Surfaces of friction-type connecting (unless primed with Employer approved primer)
- As noted on drawings

9. HANDLING, SHIPPING, DELIVERY AND STORAGE

9.1 Handling and Shipping

Delivery of steel shall be made in the order needed for erection. The delivery sequence for the fabricated steel, unless otherwise noted in the Contract Documents, shall be prior the schedule of the particular steelwork erection and as notified by Employer to Contractor.

All bolts, washers and nuts shall be packaged and delivered in rigid, weatherproof containers.

The contractor shall ensure that all steel and its coatings are protected from any damage caused by handling, storage or shipping before receipt.

The contractor shall ensure that adequate protection is provided for threads on sag rods and any other threaded components to prevent damage during shipping and handling.

9.2 Delivery

All materials and documentation will be inspected immediately upon receipt to determine that all items included in the Bill of Materials have been supplied, to assure that all documentation has been received and to check for any damage. Employer reserves the right to reject all damaged or below-quality material or documentation.

9.3 Storage and Handling

All necessary supervision, chains, timber dunnage, tie-down equipment, flags, permits and the like shall be provided to safely and carefully complete transportation of structural steel.

Structural steel for erection shall be stacked clear of the ground and be self-draining.

Whenever possible, wooden supports in contact with finished items during transport and storage shall be covered with a soft material such as rubber to minimize damage to the coating. Projections from steelwork such as lugs and splice plates which may suffer damage during transport are required to be reinforced with a timber packing piece.

Hessian fabric or carpet shall be used between the painted items and holding down chains to prevent damage to the paint coating, during transportation.

All steelwork that has been finish coated shall be handled with care so as to preserve the coating in the best practical condition. Slings used for the handling of the finished steelwork shall be covered

with soft material.

All small loose items including bolts, nuts and washers shall be supplied in grit-free containers and shall be kept in water-tight storage. Each box shall be identified with a securely fastened metal tag bearing its identification number and the approximate reference of its intended use.

Any damage sustained during handling shall be reported to the Employer immediately after it is discovered.

10. INSPECTION AND TESTING

10.1 General

Unless approved in writing beforehand by the Employer, all non-destructive testing (NDT) of welding shall be carried out by an independent authorized Testing Agency/Laboratory. All NDT operators shall be experienced and shall be trained and certified.

The contractor shall comply with the Project Quality Requirements for specific QA/QC requirements

10.2 Material

Ensure that materials meet the requirements of Section 4 and Section 5 of this Specification prior to installation.

10.3 Dimensional Inspection

The QA/QC procedures shall identify a systematic process of dimension control and verification. The dimensions of all structural components shall be inspected and verified against shop detail drawings as complying with the tolerance limits specified in this Specification.

10.4 Welding Inspection

Edge preparations and alignments of welded joints shall be inspected to ensure they comply with the requirements of AWS D1.1.

The locations of the weld lengths to be examined/inspected shall be selected by the Employer unless otherwise specified.

Types and locations of all test and weld defects found in the work shall be recorded and work required and performed to correct the defects shall be recorded.

The NDT for welding shall be comply to AWS B1.10M/B1.10.

10.5 Identification Marks

The correct identification of each completed structural component, as given by its member piece mark number, orientation etc. shall be verified with respect to Marking Plans, Shop Detail Drawings and/or Grating Arrangement Drawings.